2018-DSE
BIO

PAPER 1A

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2018

BIOLOGY PAPER 1

8:30 am – 11:00 am (2 hours 30 minutes)

This paper must be answered in English

GENERAL INSTRUCTIONS

(1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.

(2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book **B**.

(3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book **B**. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

(1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.

(2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.

(3) All questions carry equal marks.

(4) **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.

(5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

(6) No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

There are 36 questions in this section.
The diagrams in this section are NOT necessarily drawn to scale.

1. Which of the following processes involves enzymes on cell membranes?

   A. excretion of carbon dioxide by the lungs
   B. transport of water along the xylem vessel
   C. Calvin cycle in the chloroplasts of plant cells
   D. digestion of carbohydrates in the small intestine

Directions: Questions 2 to 4 refer to an experiment on the enzyme catalase, which speeds up the breakdown of hydrogen peroxide to release oxygen. John added a 1 cm^3 cube of pig liver to a boiling tube containing 5 mL hydrogen peroxide solution. Gas bubbles were released and he used a glowing splint to test the gas. He repeated the experiment with beef, potato and apple. The results are shown below:

Tissue	*Rate of bubbles released*	*Glowing splint re-lit*
Pig liver	Moderate	Yes
Beef	Moderate	Yes
Potato	Slow	Yes
Apple	Slow	Yes

2. Which of the following statements is an observation of the experiment?

   A. These tissues contained catalase.
   B. Oxygen gas was released in the reaction.
   C. The gas released re-lit the glowing splint.
   D. Animal tissues had more catalase than plant tissues.

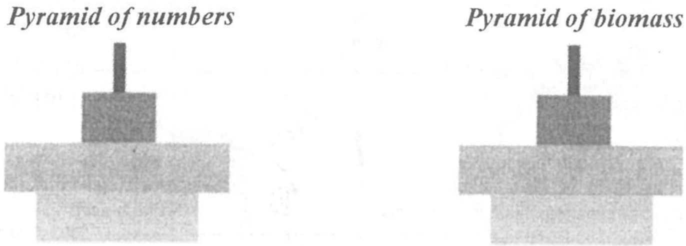
3. When the release of gas bubbles had stopped, John added more hydrogen peroxide solution to the boiling tubes. Which of the following combinations correctly shows the expected result and the explanation of this additional experiment?

	Expected result	*Explanation*
A.	Gas bubbles released	Catalase is specific in its action.
B.	Gas bubbles released	Catalase is reusable.
C.	No gas bubbles	Catalase has been used up.
D.	No gas bubbles	Catalase is denatured.

4. In order to prove that hydrogen peroxide is the substrate of this enzymatic reaction, which of the following steps should be used as a control?

   A. Repeat the experiment using water and the tissues.
   B. Repeat the experiment using water and boiled tissues.
   C. Repeat the experiment using hydrogen peroxide but no tissues.
   D. Repeat the experiment using hydrogen peroxide and boiled tissues.

5. The diagrams below show the pyramids of numbers and biomass of a food chain:



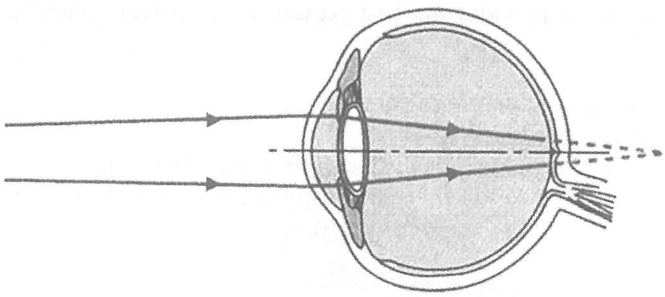
Which of the following is most likely to be the producer in this food chain?

A. rice
B. trees
C. grass
D. phytoplankton

6. Protecting sharks in the wild is important for maintaining the ecological balance of the marine ecosystem because

A. sharks are an endangered species.
B. shark fin is a popular dish for banquets.
C. the dead bodies of sharks are an important food source for decomposers.
D. sharks are top predators that regulate the population sizes of other consumers.

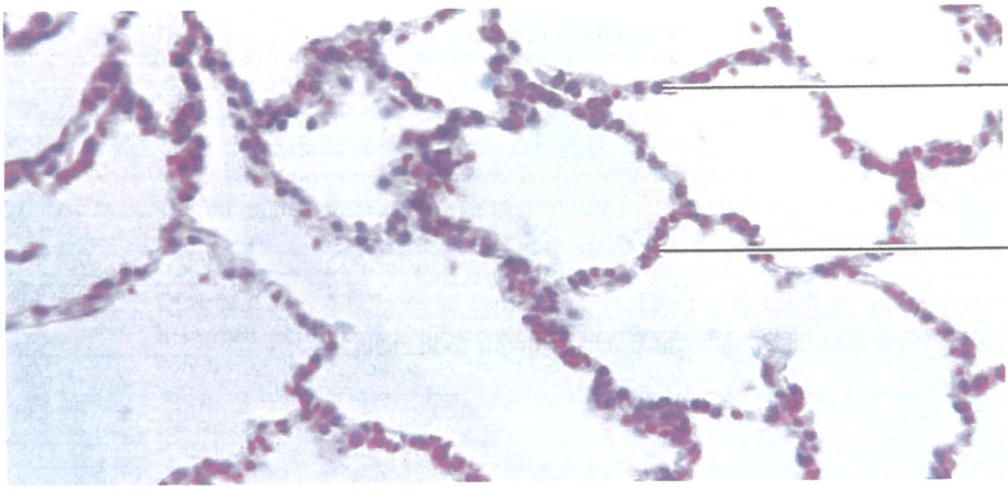
7. Below is a ray diagram of a common eye defect:



Which of the following combinations correctly identifies the eye defect and type of lens to be worn to remedy it?

	Eye defect	*Type of lens*
A.	Long-sightedness	Convex lens
B.	Long-sightedness	Concave lens
C.	Short-sightedness	Convex lens
D.	Short-sightedness	Concave lens

8. The photomicrograph below shows a section of human lung:



With reference to the structures shown in the photomicrograph, which of the following are adaptive features for gas exchange?

(1) presence of water film
(2) short diffusion distance
(3) rich supply of blood capillaries

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

9. Which of the following statements provides the best reason for classifying unicellular organisms into domain Bacteria and Archaea?

A. Archaea are more ancient than bacteria.
B. Archaea are smaller than bacteria.
C. The DNA sequences of archaea are distinct from those of bacteria.
D. The compositions of the cell wall and cell membrane of archaea are different from those of bacteria.

10. Which of the following statements best explains why vaccination against flu is administered annually?

A. The flu virus is constantly mutating.
B. The antibodies against flu virus only last for one year.
C. The flu vaccine is not very effective because it is made from a weakened virus.
D. When memory cells encounter the flu vaccine again, a secondary response is triggered.

11. Which of the following processes takes place at the inner membrane of mitochondria?

A. glycolysis
B. conversion of pyruvate to acetyl CoA
C. Krebs cycle
D. oxidative phosphorylation

Directions: Questions 12 and 13 refer to the information below. Kathy had two pure-bred cats, one had long white fur while the other had short black fur. It is known that fur length and fur colour are controlled by two different genes respectively. The two cats gave birth to four kittens which had long black fur.

12. Which of the following conclusions can be drawn based on the above case?

(1) Long fur is dominant over short fur.
(2) The four kittens have the same genotype for fur length and fur colour.
(3) The genes controlling fur length and fur colour are located on different chromosomes.

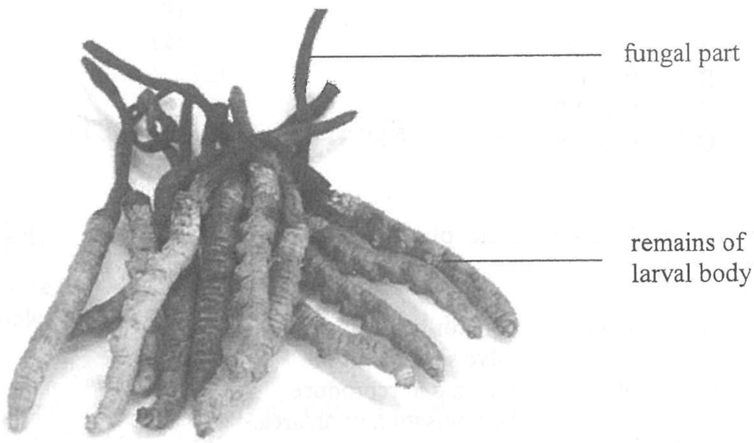
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

13. After the kittens had grown up, they interbred and gave birth to the second filial generation (F_2). Among the F_2 kittens, there was one with short white fur. Which of the following processes mostly likely contributed to the occurrence of this new phenotype?

(1) Mutation
(2) Random fertilisation of gametes
(3) Independent assortment of chromosomes

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

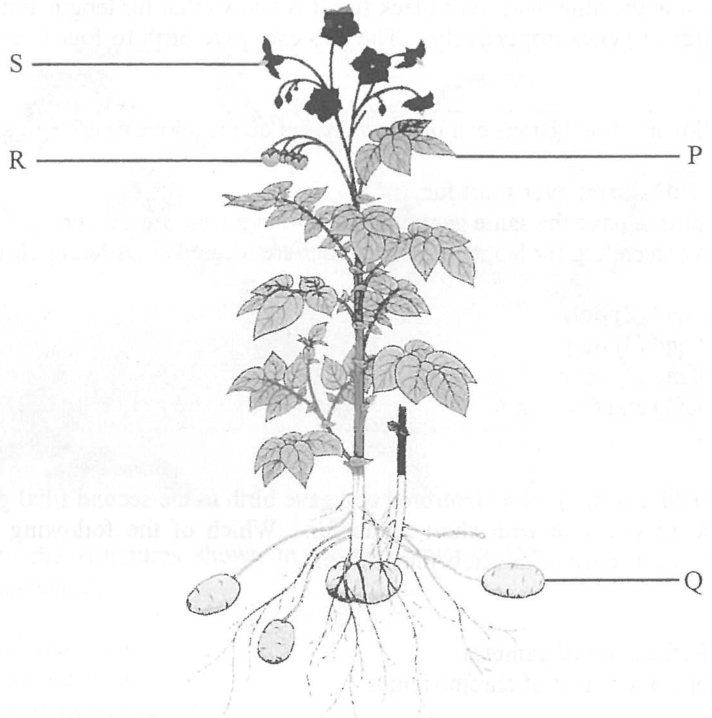
14. Caterpillar fungus is a kind of Chinese herbal medicine. When the spores of this fungus land on the larvae of moths, the spores will germinate inside the body of the larvae and grow out of their heads, as shown in the photograph below:



Which of the following best describes the role of this fungus?

A. parasite
B. predator
C. producer
D. consumer

Directions: Questions 15 to 17 refer to the following diagram, which shows the structures of a potato plant:



15. Which of the following labelled structure(s) is / are involved in the reproduction of this potato plant?

    A. Q only
    B. S only
    C. R and S only
    D. Q, R and S only

16. Which of the following labelled structures contain cells with different genotypes?

    A. P and Q
    B. P and S
    C. Q and R
    D. R and S

17. Farmers usually grow potato plants by vegetative propagation. This is probably because vegetative propagation

    A. produces tubers for harvesting.
    B. does not involve seed dispersal.
    C. takes a shorter time to reproduce.
    D. allows rapid colonisation of an area.

18. Some people claim that we should chew food for a longer time before swallowing. Which of the following statements about this claim is ***incorrect***?

    A. This stimulates the secretion of saliva.
    B. This moistens the food for easier swallowing.
    C. This provides enough time for digestion of starch into glucose.
    D. This increases the surface area of the food for chemical digestion.

Directions: Questions 19 to 21 refer to the diagrams below. Diagram I shows the reflex arc of the knee jerk reflex while diagram II shows three types of neurones:

Diagram I

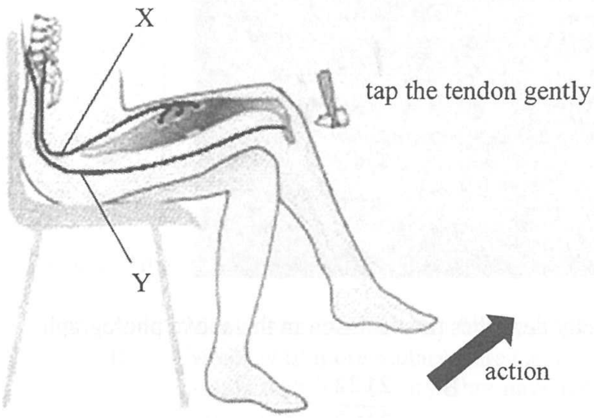
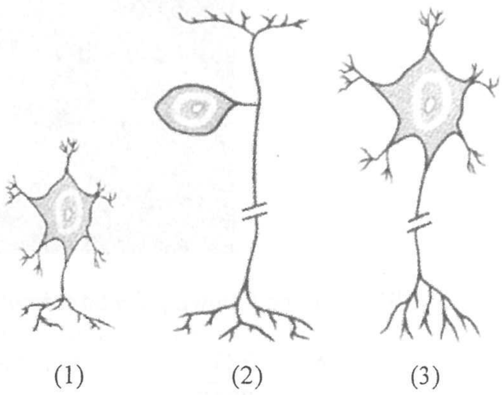


Diagram II



19. The effector in the reflex arc in Diagram I is

A. a flexor because its response bends the limb.
B. an extensor because its response straightens the limb.
C. a flexor because it shortens to bring about the movement.
D. an extensor because it lengthens to bring about the movement.

20. Which of the following combinations correctly identifies the types of neurones to which X and Y belong?

	X	*Y*
A.	(1)	(3)
B.	(2)	(1)
C.	(2)	(3)
D.	(3)	(2)

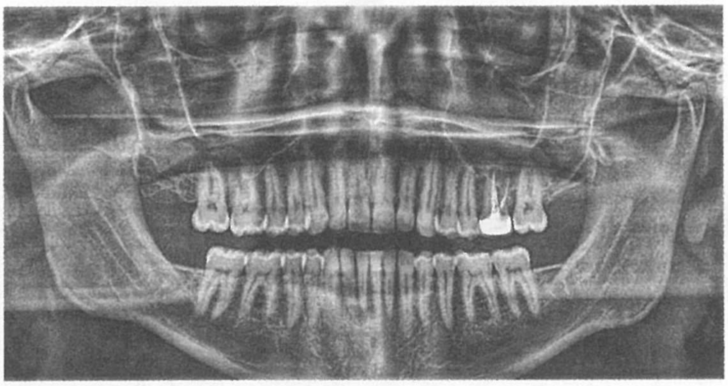
21. Another neural pathway allows the man to feel the tapping action. Which of the following parts should this pathway connect to?

A. cerebellum
B. spinal cord
C. cerebral cortex
D. medulla oblongata

22. Which of the following combinations correctly identifies the distribution of grey matter and white matter in the cerebrum and spinal cord?

	Inner part of the cerebrum	*Inner part of the spinal cord*
A.	white matter	grey matter
B.	white matter	white matter
C.	grey matter	grey matter
D.	grey matter	white matter

23. The following picture shows an X-ray photograph of the dentition of a person:

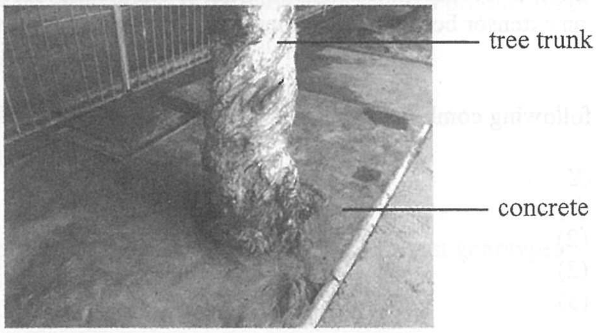


Which of the following dental formulae correctly describes the dentition in the above photograph?

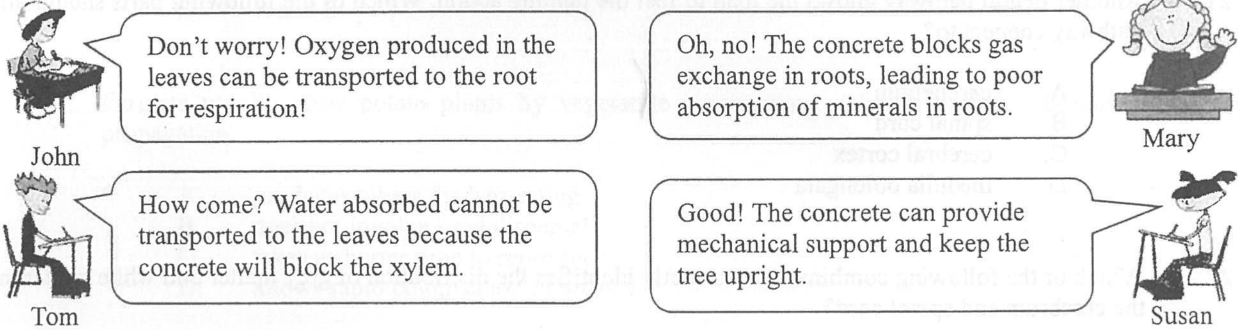
A. $\frac{2212}{2212}$ B. $\frac{2122}{2122}$

C. $\frac{2131}{2131}$ D. $\frac{2113}{2113}$

24. The following photograph shows a tree with roots covered by concrete:



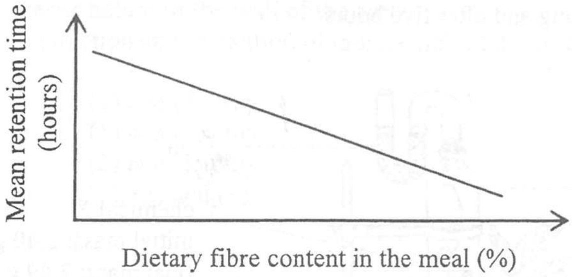
Four students have expressed their views about this:



Whose view is correct?

A. John's view
B. Mary's view
C. Tom's view
D. Susan's view

25. The graph below shows the relationship between the dietary fibre content of a meal and the mean retention time (i.e. the duration for which the undigested materials stay in the large intestine) in the human body:



Which of the following can be deduced from the graph?

A. Meals with more dietary fibre can increase the bulk of the faeces.
B. Meals with more dietary fibre can hold more water, and so soften faeces better.
C. Meals with less dietary fibre have more nutrients and a longer time is required for complete absorption.
D. Meals with less dietary fibre result in harder faeces due to an increased time for water absorption.

26. After vigorous exercise, the blood lactic acid concentration of an athlete increases. Which of the following word equations correctly shows the process that leads to the formation of lactic acid?

A. glucose → lactic acid
B. glucose → lactic acid + water
C. glucose → lactic acid + carbon dioxide
D. glucose + oxygen → lactic acid + carbon dioxide

27 'Lock and Key' is a scientific model which is a selective representation used to explain that enzymes

A. are biological catalysts.
B. are specific in action.
C. are protein in nature.
D. are required in small amounts.

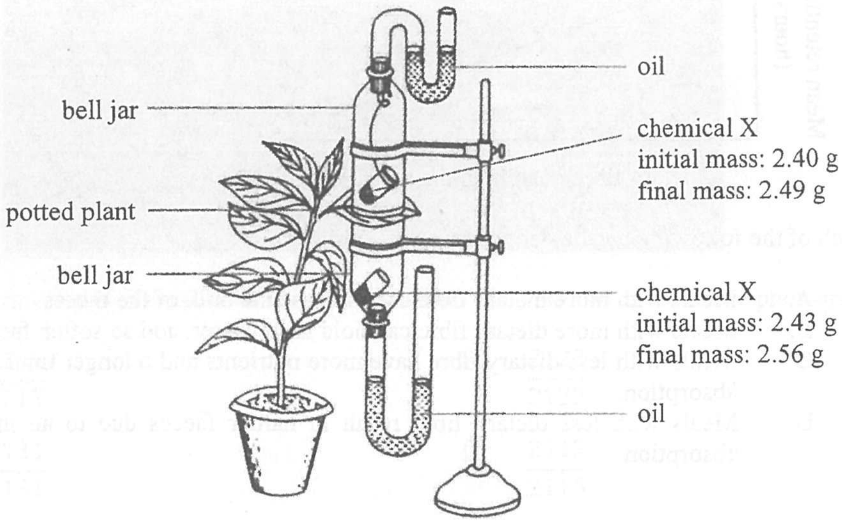
28. The DNA model proposed by Watson and Crick leads to the understanding of how

A. organisms store genetic codes.
B. organisms share a common ancestor.
C. cells produce instructions for protein synthesis.
D. cells can pass genetic information to the next generation.

29. Which of the following substances contribute(s) most to the increase in biomass of a plant?

A. water
B. oxygen
C. minerals
D. carbon dioxide

Directions: Questions 30 and 31 refer to the set-up below. The set-up consists of two bell jars placed one above the other with the leaf of a potted plant in between. Chemical X was placed into the jars to absorb water vapour. The whole set-up was made air-tight. The masses of chemical X in the two jars were measured at the beginning and after five hours.



30. The change in mass of chemical X was mainly caused by

A. water uptake by the plant.
B. water loss in transpiration.
C. water produced in respiration.
D. water consumed by photosynthesis.

31. Which of the following conclusions can be drawn from the results?

A. Water absorption by the root is mainly driven by the lower epidermis of the leaf.
B. There are more stomata at the lower epidermis than the upper epidermis of the leaf.
C. The respiration rate is higher than the photosynthetic rate during the experiment.
D. The photosynthetic rate of the upper layer of the leaf is higher than that of the lower layer.

32. Organisms P and Q are found in the same local habitat. Their population sizes have continued to grow in the last few years. Which of the following statements best describes organisms P and Q?

A. They are heterotrophs.
B. They are top predators.
C. They have different niches.
D. They have different predators.

33. Which of the following statements about primary succession and secondary succession is correct?

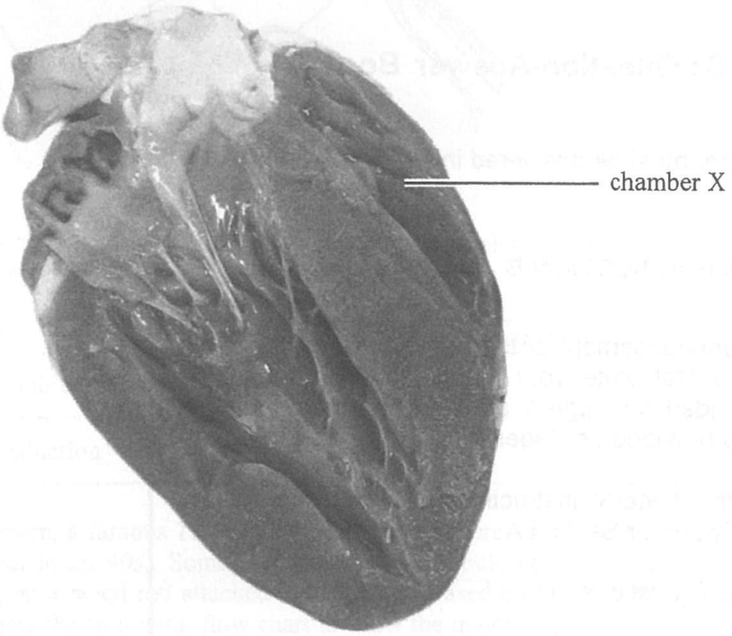
A. Primary succession is always followed by secondary succession.
B. Primary succession always starts with a barren area while secondary succession does not.
C. Secondary succession always ends with a climax community while primary succession does not.
D. Secondary succession always ends with a climax community while primary succession ends with a pioneer community.

34. Which of the following contribute to the continuous blood flow in the aorta?

(1) pumping action of the heart
(2) elastic nature of the wall of the aorta
(3) contraction and relaxation of muscle wall of the aorta

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

35. The following photograph shows a dissected pig heart:



Which of the following descriptions about chamber X is correct?

A. It receives blood from the pulmonary vein.
B. It pumps out blood to the aorta.
C. It receives blood from the vena cava.
D. It pumps out blood to the pulmonary artery.

36. Which of the following processes mainly involves osmosis?

A. movement of water along the xylem in plants
B. movement of water vapour out of stomata in plants
C. movement of water from tissue fluid to capillaries in humans
D. movement of water from tissue fluid to lymph capillaries in humans

END OF SECTION A
Go on to Question-Answer Book B for questions on Section B